

Comparing Classical and Quantum-Classical Hybrid Neural Networks for Binary MNIST Classification: A Small Ablation Study

Abstract

This study compares four neural network configurations on a binary MNIST classification task (digits 0 versus 1): a fully connected network, a convolutional network, a convolutional network with a parameter-matched classical control head, and a convolutional network with a four-qubit variational quantum circuit head. The classical control head and the quantum head were both constructed with 25 trainable parameters so that any difference in performance could be attributed to the quantum operation rather than to the addition of extra trainable parameters. All four models were trained for five independent runs each, using the same five random seeds, with early stopping and identical optimization settings across models. All four trained models exceeded a 53.66% majority-class baseline by a wide margin, reaching mean test accuracies above 99.9%. A paired Wilcoxon signed-rank test comparing the classical control model and the quantum model found no statistically distinguishable difference in accuracy, F1 score, Matthews correlation coefficient, or recall (p between 0.25 and 0.375, $n = 5$). The quantum model was the only configuration to produce false negatives in this study and required substantially longer training time under classical simulation than the other three models.

1. Introduction

Convolutional neural networks are a standard approach for image classification tasks of this kind, and variational quantum circuits have been proposed as an additional processing component within otherwise classical architectures. A common gap in reports of hybrid quantum-classical models is the absence of a classical control matched in parameter count to the quantum component, which makes it difficult to determine whether an observed change in performance comes from the quantum operation itself or simply from the addition of extra trainable parameters. This study addresses that gap directly, using a small, binary version of the MNIST handwritten digit dataset restricted to digits 0 and 1.

Four models were trained as part of an ablation chain: a majority-class baseline, a fully connected network, a convolutional network, and two variants of that convolutional network in which the final processing block was replaced by either a classical control head or a quantum head, both constructed with exactly 25 trainable parameters. The task was kept deliberately small, using four qubits, two variational circuit layers, and CPU-based classical simulation, so that the full chain, including five independent training runs per model, could be executed without specialized hardware.

The aim of this study is not to demonstrate an advantage of quantum computation over classical computation. It is to test, under matched conditions, whether replacing a small classical processing block with an equally sized quantum block changes classification performance on this task, and to report what differences do appear, whether in accuracy, error type, or training behavior.

2. Related Background

Convolutional networks process images through learned local filters, which is the reason a convolutional model was included alongside a plain fully connected network in this study. Variational quantum circuits used inside otherwise classical models, commonly described as hybrid quantum-classical models, typically encode classical input values as gate rotation angles, process them through a sequence of trainable single-qubit rotations and entangling operations, and produce an output by measuring expectation values that are then passed to a small classical output layer. This general structure, rather than any specific published architecture, is the basis for the quantum head used in this study.

No systematic literature review was conducted for this study. This section is limited to a general description of the type of architecture used rather than a survey of prior published results, so that no claim in this report is attributed to a source that was not directly consulted.

3. Methodology

3.1 Dataset

The dataset used was MNIST, restricted to the digits 0 and 1. The official MNIST training and test sets were kept separate throughout the study; no image from the official test set was used for training, validation, or any hyperparameter decision. Restricting the official training set to digits 0 and 1 produced a pool of 12,665 images, split 80/20 into a training set and a validation set using a stratified split so that both subsets preserved the class ratio of the full pool. The official test set, restricted the same way, produced 2,115 images and was not used until the final evaluation of each trained model. Table 1 summarizes the resulting subsets. Pixel normalization statistics (mean 0.1215, standard deviation 0.3010) were computed from the training split only and applied unchanged to the validation and test sets. Digit 1 was treated as the positive class and digit 0 as the negative class throughout.

Subset	Images	Notes
Training	10,132	Stratified 80% split of the binary training pool
Validation	2,533	Stratified 20% split of the binary training pool
Test (official)	2,115	Held out from the official MNIST test set; used only for final evaluation
Binary training pool	12,665	5,923 zeros (46.77%), 6,742 ones (53.23%)

Table 1. Composition of the dataset splits used in this study.

3.2 Model Architectures

Table 2 summarizes the four trained architectures. All four models take a 28x28 grayscale image as input and produce a single raw logit, converted to a class prediction with a sigmoid function and a threshold of 0.5.

The fully connected network (NN) flattens each image into a 784-element vector and passes it through two hidden layers (128 and 64 units, ReLU activations) to a single output unit, giving 108,801 trainable parameters. It served as a standalone baseline and was not parameter-matched to the other three models.

The convolutional network (CNN) and both of its head variants share an identical backbone: two convolution and max-pooling blocks (1 to 16 channels, then 16 to 32 channels, each followed by 2x2 max pooling), a flattening step, and a linear layer reducing the resulting 800 values to a 4-dimensional feature vector. This backbone contains 8,004 trainable parameters and was reused without modification across the plain CNN, the classical control model, and the quantum model, so that the only architectural difference between these three models is the processing applied to the 4-dimensional feature vector.

The plain CNN head is a single linear layer (4 to 1), adding 5 parameters. The classical control head applies a linear layer (4 to 4) with a ReLU activation followed by a second linear layer (4 to 1), adding 25 parameters. This head exists specifically to match the parameter count of the quantum head described in Section 3.3, so that the comparison between the classical control model and the quantum model isolates the effect of the quantum operation itself rather than the effect of added processing capacity.

Model	Backbone	Head	Head params	Total params
NN	None (flatten only)	784-128-64-1, ReLU	-	108,801
CNN	Conv(1-16)-Pool-Conv(16-32)-Pool-Linear(800-4)	Linear(4-1)	5	8,009
CNN + Classical	Same backbone as CNN	Linear(4-4)-ReLU-Linear(4-1)	25	8,029
CNN + Quantum	Same backbone as CNN	Scale(4) + RY/RZ circuit(16) + Linear(4-1)(5)	25	8,029

Table 2. Summary of the four trained model architectures.

3.3 Quantum Circuit Design

The quantum head begins with a learnable per-feature scaling parameter (4 values, no bias term, no mixing across features) applied to the CNN backbone's 4 output features, followed by a $\pi \cdot \tanh(x)$ transformation bounding the scaled values to the range $(-\pi, \pi)$. These four bounded values are encoded onto four qubits using RY rotations. The circuit then applies two identical variational layers, each consisting of an RY and an RZ rotation on every qubit, followed by a ring of CNOT gates connecting qubit i to qubit $(i+1) \bmod 4$, closing the ring from qubit 3 back to qubit 0. After the second layer, the expectation value of the Pauli Z operator is measured on each of the four qubits, producing four output values that are passed to a final linear layer (4 to 1).

The RY and RZ rotations across the two variational layers give 16 trainable quantum parameters (4 qubits x 2 rotations x 2 layers). Combined with the 4 scaling parameters and the 5 parameters of the final linear layer, the quantum head totals 25 trainable parameters, matching the classical control head exactly.

Quantum weight initialization used values drawn from a normal distribution with mean 0 and standard deviation 0.1, rather than the He initialization used for the classical layers, since He initialization has no established basis for periodic rotation angles that are not followed by a ReLU nonlinearity. This initialization was applied by overwriting the layer's weights directly after construction, because passing the initialization through PennyLane's `init_method` argument produced a `torch.normal` argument-parsing error in the environment used for this study.

Before use inside the full circuit, individual gate behaviors were verified in isolated tests. Two early attempts to demonstrate that an RZ rotation on one qubit affects a measurement on an entangled partner qubit gave results showing no effect, because both attempts measured the Pauli Z operator on the partner qubit, which does not reveal a phase-only rotation. Repeating the test using a Hadamard-RZ-Hadamard sequence on a single qubit correctly reproduced the expected cosine relationship between the RZ angle and the resulting Z expectation value. This was treated as an implementation and understanding check prior to constructing the full circuit, not as a result of the study itself.

3.4 Training Protocol

All four models were trained using the Adam optimizer with a fixed learning rate of 0.001, binary cross-entropy loss applied to the raw logit, a batch size of 32, and early stopping on validation loss with a patience of 5 epochs and a maximum of 30 epochs, restoring the weights from the epoch with the lowest validation loss. Each model was trained for five independent runs using the same five random seeds (42, 123, 2024, 7, 99), applied identically across all four models and covering the Python, NumPy, and PyTorch random states. DataLoader workers were set to 0 and deterministic algorithm execution was enabled for all runs.

3.5 Pilot Testing

Before the five official runs of the quantum model, a single seed (42) was used to test the full pipeline and check for implementation errors not necessarily visible in final accuracy figures. This testing identified two issues in the initial quantum layer implementation. First, the PennyLane lightning.qubit backend produced a reshape error when passed a batch of inputs through the PyTorch integration layer; the circuit was switched to the default.qubit backend, which handled batched execution correctly. This is reported as a simulator backend substitution rather than a result of the study. Second, the circuit's input indexing initially selected sample i from the batch rather than feature i across the batch, which caused a runtime error for any batch size greater than 1 and would have forced an incorrect workaround had it not been caught before the official runs.

After this correction, a gradient check on one training batch showed nonzero gradients on all parameters, including a gradient norm of 0.227761 on the 16 quantum rotation parameters, and a batch-size timing check (8, 16, 32, 64) showed execution time remaining approximately constant after an initial warm-up call, consistent with batched rather than per-sample execution. A five-epoch pilot training run reached 73.6% training accuracy after the first epoch and 100% by the fifth epoch, compared with the classical models, which typically exceeded 99% training accuracy already after the first epoch.

4. Experimental Setup

All models were trained on CPU only, consistent with the small scale of the task; the four-qubit circuit has a state space of size 16. Software versions used were PyTorch 2.11.0+cpu, Torchvision 0.26.0+cpu, NumPy 2.0.2, PennyLane 0.45.1, and scikit-learn 1.6.1, run in a Google Colab environment with data and results stored on Google Drive. The quantum device used for the official runs was PennyLane's default.qubit simulator with analytic, non-sampled expectation value computation.

5. Results

5.1 Overall Performance

Table 3 reports the mean and standard deviation of nine metrics across the five runs of each model. The majority-class baseline, which always predicts digit 1, reached 53.66% test accuracy. All four trained models exceeded this baseline by a wide margin, reaching mean test accuracies above 99.9%. Differences between the trained models were small in absolute terms: the fully connected network reached 99.95% mean accuracy with negligible variance across seeds, the plain CNN and the classical control model both reached 99.96% mean accuracy, and the quantum model reached 99.92% mean accuracy with the highest standard deviation among the four trained models.

Model	Accuracy	Precision	Recall	Specificity	F1
NN	0.9995 ± 0.0000	0.9991 ± 0.0000	1.0000 ± 0.0000	0.9990 ± 0.0000	0.9996 ± 0.0000
CNN	0.9996 ± 0.0004	0.9993 ± 0.0007	1.0000 ± 0.0000	0.9992 ± 0.0009	0.9996 ± 0.0004
CNN + Classical	0.9996 ± 0.0002	0.9993 ± 0.0004	1.0000 ± 0.0000	0.9992 ± 0.0005	0.9996 ± 0.0002
CNN + Quantum	0.9992 ± 0.0005	0.9993 ± 0.0007	0.9993 ± 0.0007	0.9992 ± 0.0009	0.9993 ± 0.0005

Table 3a. Mean $\pm$ standard deviation across five runs (accuracy, precision, recall, specificity, F1).

Model	Balanced accuracy	MCC	ROC-AUC	PR-AUC
NN	0.9995 ± 0.0000	0.9990 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
CNN	0.9996 ± 0.0004	0.9992 ± 0.0008	1.0000 ± 0.0000	1.0000 ± 0.0000
CNN + Classical	0.9996 ± 0.0002	0.9992 ± 0.0004	1.0000 ± 0.0000	1.0000 ± 0.0000
CNN + Quantum	0.9992 ± 0.0005	0.9985 ± 0.0011	1.0000 ± 0.0000	1.0000 ± 0.0000

Table 3b. Mean $\pm$ standard deviation across five runs (balanced accuracy, MCC, ROC-AUC, PR-AUC).

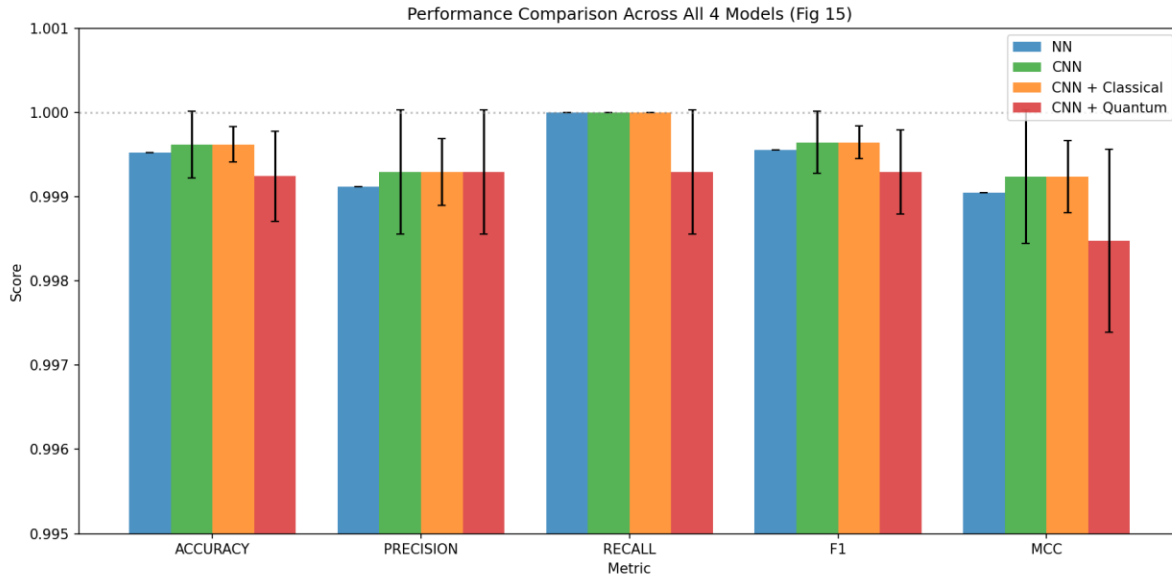


Figure 1. Mean and standard deviation of five metrics across the four trained models (y-axis truncated to 0.995-1.001 to show separation near ceiling).

5.2 Computational Cost

Table 4 reports parameter counts, average training time, and the average epoch at which the best validation loss occurred for each model. Training time increased from the fully connected network (41.0 seconds) through the CNN (108.6 seconds) and the classical control model (229.5 seconds) to the quantum model (508.0 seconds), despite the CNN-family models having far fewer parameters than the fully connected network. The quantum model required roughly twelve times the average training time of the fully connected network while using about thirteen times fewer parameters, indicating that the added time reflects the cost of simulating the quantum circuit rather than parameter count. This is reported strictly as a cost of classical simulation on the hardware used in this study and should not be read as a statement about the speed of quantum computation on physical hardware. The quantum model also required more epochs on average to reach its best validation loss (23.2) than the classical control model (17.6), the CNN (5.0), or the fully connected network (3.4).

Model	Parameters	Avg. training time (s)	Avg. best epoch	Avg. total epochs
NN	108,801	41.0	3.4	8.4
CNN	8,009	108.6	5.0	10.0
CNN + Classical	8,029	229.5	17.6	20.8
CNN + Quantum	8,029	508.0	23.2	27.4

Table 4. Parameter counts and training time by model, averaged over five runs.

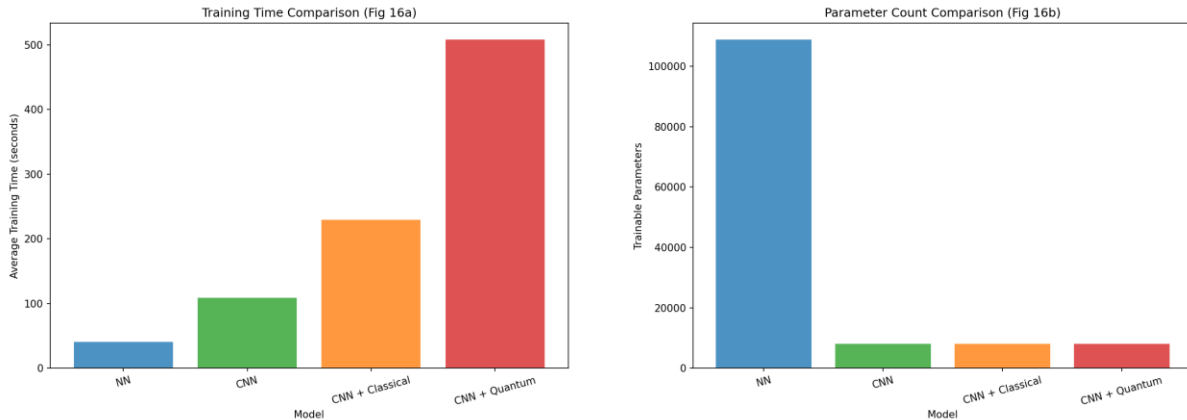


Figure 2. Average training time (left) and trainable parameter count (right) by model.

5.3 Training Curves

Figures 3 through 6 show training and validation loss and accuracy across all five runs of each model. The fully connected network and the plain CNN reached low loss within the first few epochs for every seed. The classical control model shows the two convergence patterns described in Section 5.5 directly: three runs drop quickly while two runs (seeds 123 and 7) decrease loss gradually over the full epoch budget. The quantum model's five runs are visually closer to one another than the classical control model's, consistent with the lower run-to-run spread in its training curves, even though its loss decreases more gradually overall and it requires more epochs to reach a comparably low validation loss.

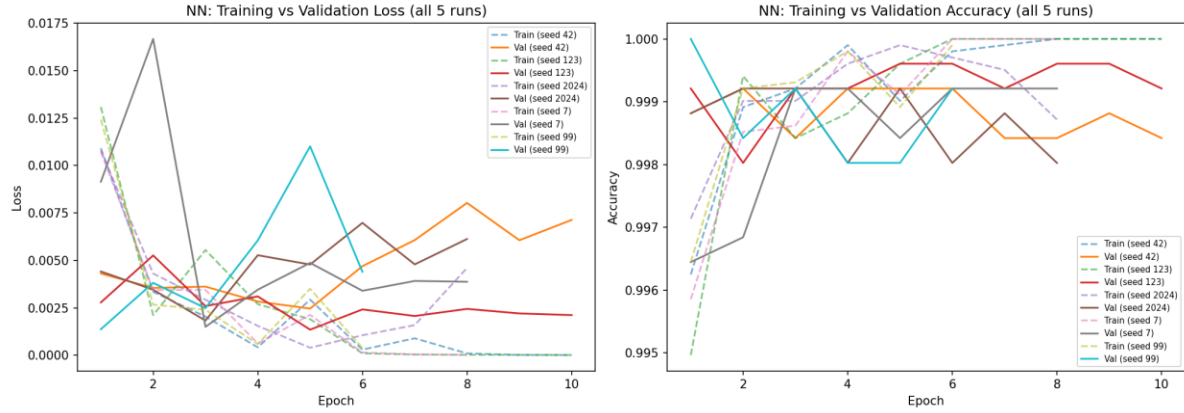


Figure 3. NN training and validation loss and accuracy, all five runs.

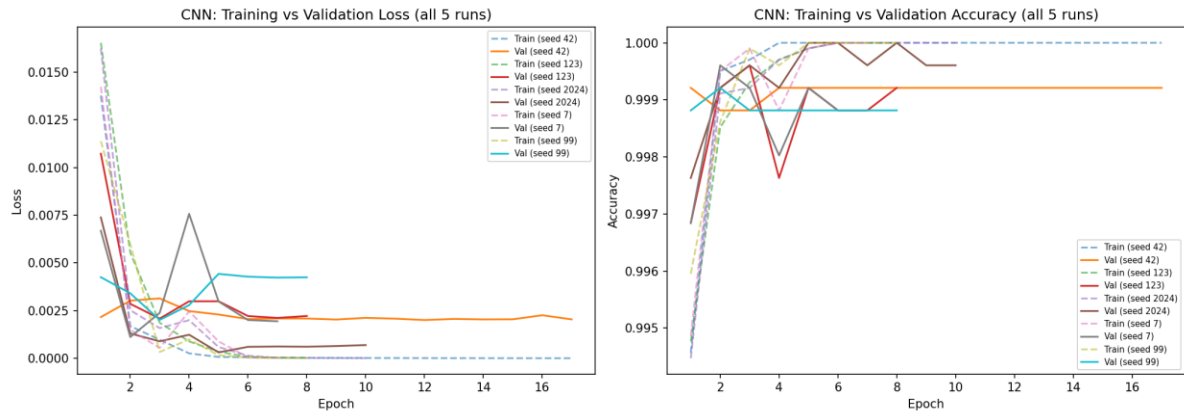


Figure 4. CNN training and validation loss and accuracy, all five runs.

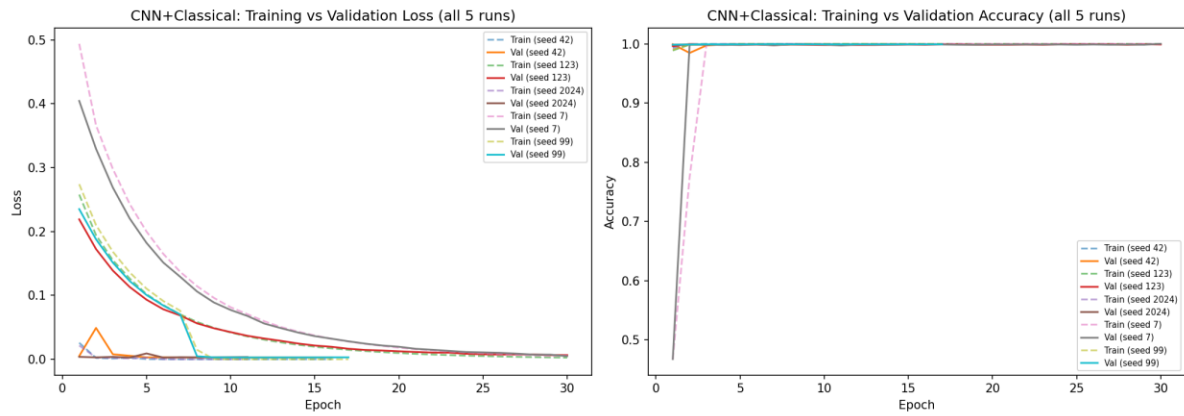


Figure 5. CNN + Classical training and validation loss and accuracy, all five runs. Seeds 123 and 7 show markedly slower loss reduction than the remaining three seeds.

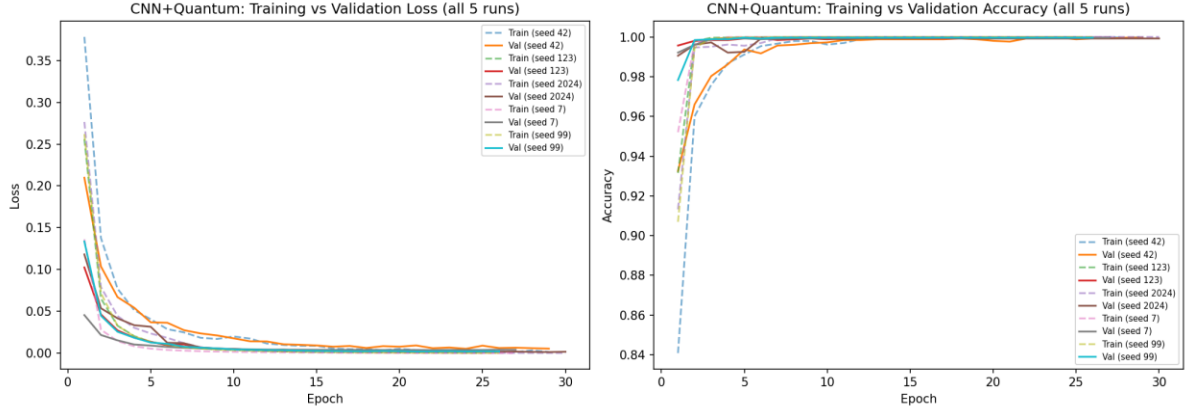


Figure 6. CNN + Quantum training and validation loss and accuracy, all five runs.

5.4 Confusion Matrices and Error Analysis

Across all five runs of the plain CNN and the classical control model, no run produced a false negative, meaning an image of digit 1 misclassified as digit 0; the small number of errors that did occur in these two models were false positives. The quantum model was the only one of the four trained models to produce false negatives, doing so in three of its five runs. Table 5 lists the confusion matrix values for each of the five quantum model runs. This is a difference in the type of error made, not only its frequency: the classical control model and the plain CNN never missed a digit 1 in this study, while the quantum model did in most of its runs, at a low but nonzero rate.

Run	TN	FP	FN	TP
Run 1 (seed 42)	978	2	1	1134
Run 2 (seed 123)	980	0	0	1135
Run 3 (seed 2024)	979	1	1	1134
Run 4 (seed 7)	980	0	2	1133
Run 5 (seed 99)	979	1	0	1135

Table 5. Confusion matrix values for the CNN + Quantum model, by run, on the 2,115-image test set.

Figure 7 shows the confusion matrix for a representative run (seed 42) of each of the four trained models, illustrating the false-negative pattern directly: the quantum model is the only one of the four with a nonzero off-diagonal entry in the digit-1 row. Full per-run confusion matrix values for all four models are listed in Appendix A.

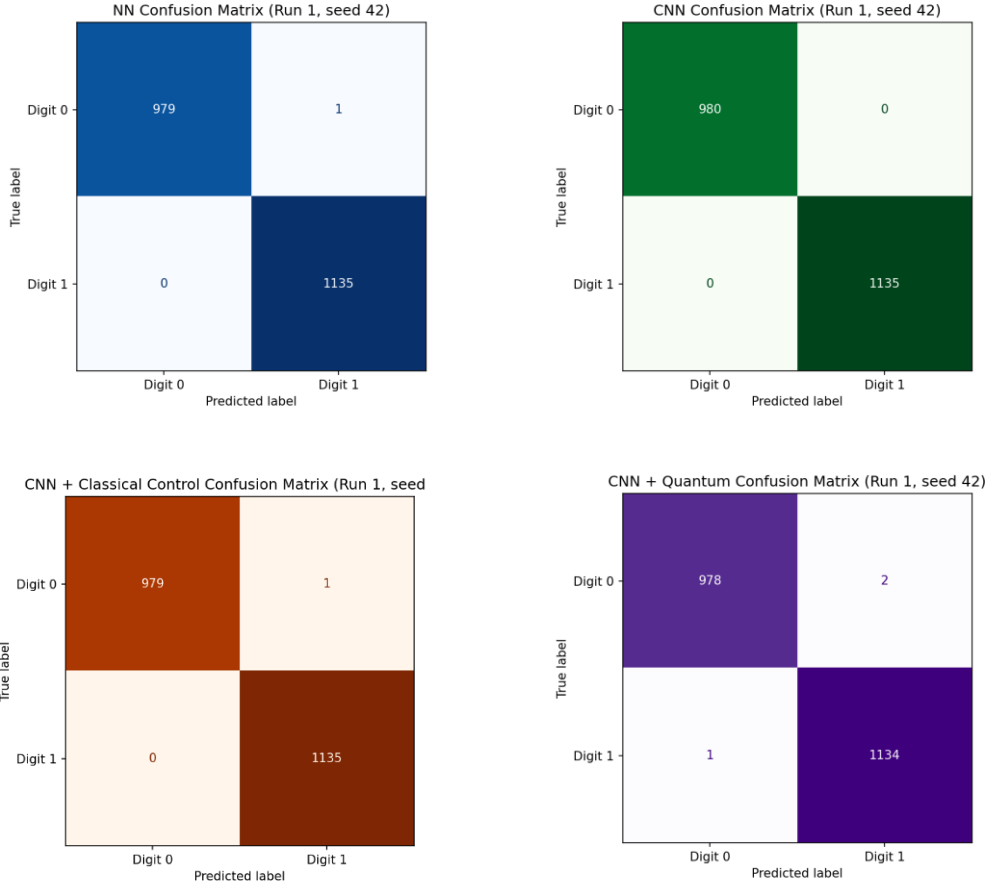


Figure 7. Confusion matrices for a representative run (seed 42) of each model: NN, CNN, CNN + Classical, CNN + Quantum (left to right).

5.5 Training Stability

The classical control model showed two distinct training patterns across its five runs, despite identical architecture, data, and hyperparameters, differing only by random seed. Three runs (seeds 42, 2024, 99) reached their best validation loss within 6 to 17 epochs, with loss decreasing quickly. Two runs (seeds 123 and 7) decreased loss much more slowly over the full 30-epoch budget without triggering early stopping; the seed 7 run recorded 46.8% training accuracy after its first epoch, close to chance level for this task, before recovering over the following epochs. All five runs nonetheless converged to closely similar final test accuracy, so this instability was visible only in the epoch-by-epoch training history, not in the final results table.

The quantum model did not show this pattern. All five of its runs required a broadly similar number of epochs, between 20 and 29, to reach their best validation loss, and no run showed near-chance accuracy at any point. In this respect, the quantum model's training behavior was more consistent across seeds than that of the classical control model, even though the classical control model reached a marginally higher mean test accuracy.

5.6 ROC Curve Comparison

Figure 8 shows the ROC curve for a representative run of each of the four trained models on the same axes, alongside a zoomed view of the top-left corner, since all four curves reach the top-left region so closely that they are difficult to distinguish at full scale. Individual ROC curves for all five runs of each model separately are provided in Appendix B; run-to-run overlap within each model was close enough that the five curves for a given model are not visually distinguishable from one another at this scale.

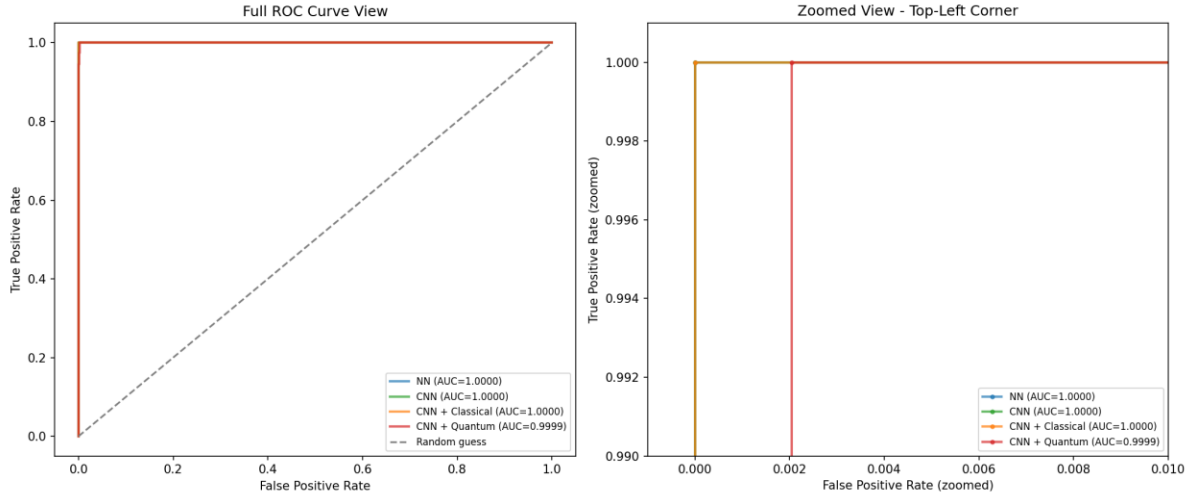


Figure 8. ROC curves for a representative run of each model (left), with a zoomed view of the top-left corner (right).

5.7 Statistical Comparison

The primary comparison of this study, the classical control model against the quantum model, was tested using a paired Wilcoxon signed-rank test across the five matched seeds. Table 6 reports the results. The test found no statistically distinguishable difference between the two models on accuracy ($p = 0.375$), F1 score ($p = 0.25$), Matthews correlation coefficient ($p = 0.25$), or recall ($p = 0.25$). The 95% confidence interval for the mean accuracy difference (classical minus quantum), computed using a Student's t interval with 4 degrees of freedom, was -0.000387 to 0.001144 , which includes zero. With five paired observations, the smallest possible two-sided Wilcoxon p -value is 0.0625 , so this test could not have reached conventional statistical significance under any outcome at this sample size; the result is reported as exploratory rather than confirmatory evidence.

Metric	Wilcoxon p-value	Mean difference (Classical - Quantum)	95% CI (t, df=4)
Accuracy	0.375	0.000378	$[-0.000387, 0.001144]$
F1 score	0.250	0.000353	not computed
MCC	0.250	0.000760	not computed
Recall	0.250	0.000705	not computed

Table 6. Paired Wilcoxon signed-rank test, CNN + Classical versus CNN + Quantum, $n = 5$.

6. Discussion

Under the conditions of this study, replacing a 25-parameter classical head with an equally sized quantum head did not produce a measurable improvement in classification accuracy, and the classical control model performed marginally better on average, though not to a statistically distinguishable degree at this sample size. This result does not indicate that the quantum circuit had no effect on model behavior. Two differences appeared between the classical control and quantum models that were not captured by the accuracy comparison alone. The quantum model was the only model to produce false negatives, suggesting a shift in the type of error the model makes rather than only its rate, though five runs are too few to determine whether this pattern would hold with additional seeds or on a different task. The quantum model also showed no training instability across seeds, in contrast to the classical control model's two unstable runs, while requiring consistently more epochs and substantially more wall-clock time under simulation to reach its best result.

Taken together, these observations suggest that on this small and comparatively easy classification task, the choice between a classical and a quantum head of matched size affected training dynamics and error type more clearly than it affected final accuracy. Because binary 0-versus-1 MNIST classification is close to ceiling performance for all four models tested, including the fully connected baseline, this task may not provide enough headroom to detect a moderate difference in representational capacity even if one exists at this scale.

7. Limitations

This study has several limitations specific to its design. The classification task, binary 0-versus-1 MNIST, is simple enough that all four trained models exceeded 99.9% mean accuracy, leaving little room to observe a meaningful separation between models regardless of architecture. The quantum circuit used four qubits and two variational layers, a configuration chosen so that all experiments could run on CPU within a practical amount of time; results may not generalize to larger circuits, different qubit counts, or different encoding schemes. All quantum computation was performed using a classical simulator rather than physical quantum hardware, and the longer training time observed for the quantum model reflects the cost of this simulation rather than a property of quantum hardware. Each model was trained for five runs, sufficient to report an exploratory statistical comparison but not enough to detect small effects with confidence, since the smallest possible two-sided Wilcoxon p-value at this sample size is 0.0625. The classical control head and the quantum head were matched by trainable parameter count, which controls for one obvious confound but does not imply that the two heads have equivalent expressive power, since a linear-ReLU-linear transformation and a variational quantum circuit are structurally different computations. No systematic literature review was conducted for this study; the background description in Section 2 is limited to a general architectural description rather than a survey of prior published results.

8. Conclusion

This study compared a majority-class baseline, a fully connected network, a convolutional network, a convolutional network with a parameter-matched classical control head, and a convolutional network with a parameter-matched quantum head, on binary 0-versus-1 MNIST classification, using five training runs per model with matched seeds. All four trained models substantially exceeded the majority baseline and

reached mean test accuracies above 99.9%. A paired statistical comparison between the classical control and quantum models found no statistically distinguishable difference in accuracy, F1 score, Matthews correlation coefficient, or recall at five runs. The quantum model was the only model to produce false negatives, converged more slowly on average, required more wall-clock time under simulation, and showed more consistent training behavior across seeds than the classical control model. Extending this comparison to a larger or more difficult classification task, where models are less likely to reach near-ceiling performance, would be a natural next step for testing whether these training-dynamics and error-pattern differences persist, and whether they translate into a measurable accuracy difference at a larger scale.

Appendix A: Full Per-Run Confusion Matrices

Table A1 lists the confusion matrix values for every individual run of all four trained models on the 2,115-image official test set, extending the single representative run shown in Figure 7 and Table 5.

Model	Run	TN	FP	FN	TP
NN	Run 1 (seed 42)	979	1	0	1135
	Run 2 (seed 123)	979	1	0	1135
	Run 3 (seed 2024)	979	1	0	1135
	Run 4 (seed 7)	979	1	0	1135
	Run 5 (seed 99)	979	1	0	1135
CNN	Run 1 (seed 42)	980	0	0	1135
	Run 2 (seed 123)	978	2	0	1135
	Run 3 (seed 2024)	979	1	0	1135
	Run 4 (seed 7)	979	1	0	1135
	Run 5 (seed 99)	980	0	0	1135
CNN + Classical	Run 1 (seed 42)	979	1	0	1135
	Run 2 (seed 123)	979	1	0	1135
	Run 3 (seed 2024)	979	1	0	1135
	Run 4 (seed 7)	980	0	0	1135
	Run 5 (seed 99)	979	1	0	1135
CNN + Quantum	Run 1 (seed 42)	978	2	1	1134
	Run 2 (seed 123)	980	0	0	1135
	Run 3 (seed 2024)	979	1	1	1134
	Run 4 (seed 7)	980	0	2	1133
	Run 5 (seed 99)	979	1	0	1135

Table A1. Confusion matrix values (TN, FP, FN, TP) for every run of every trained model.

Appendix B: Individual ROC Curves by Model

Figures B1 through B4 show the ROC curve for each of the five individual runs of each trained model, plotted on the same axes. In every case the five curves for a given model overlap closely enough to be visually indistinguishable, consistent with the near-identical ROC-AUC values reported in Table 3b.

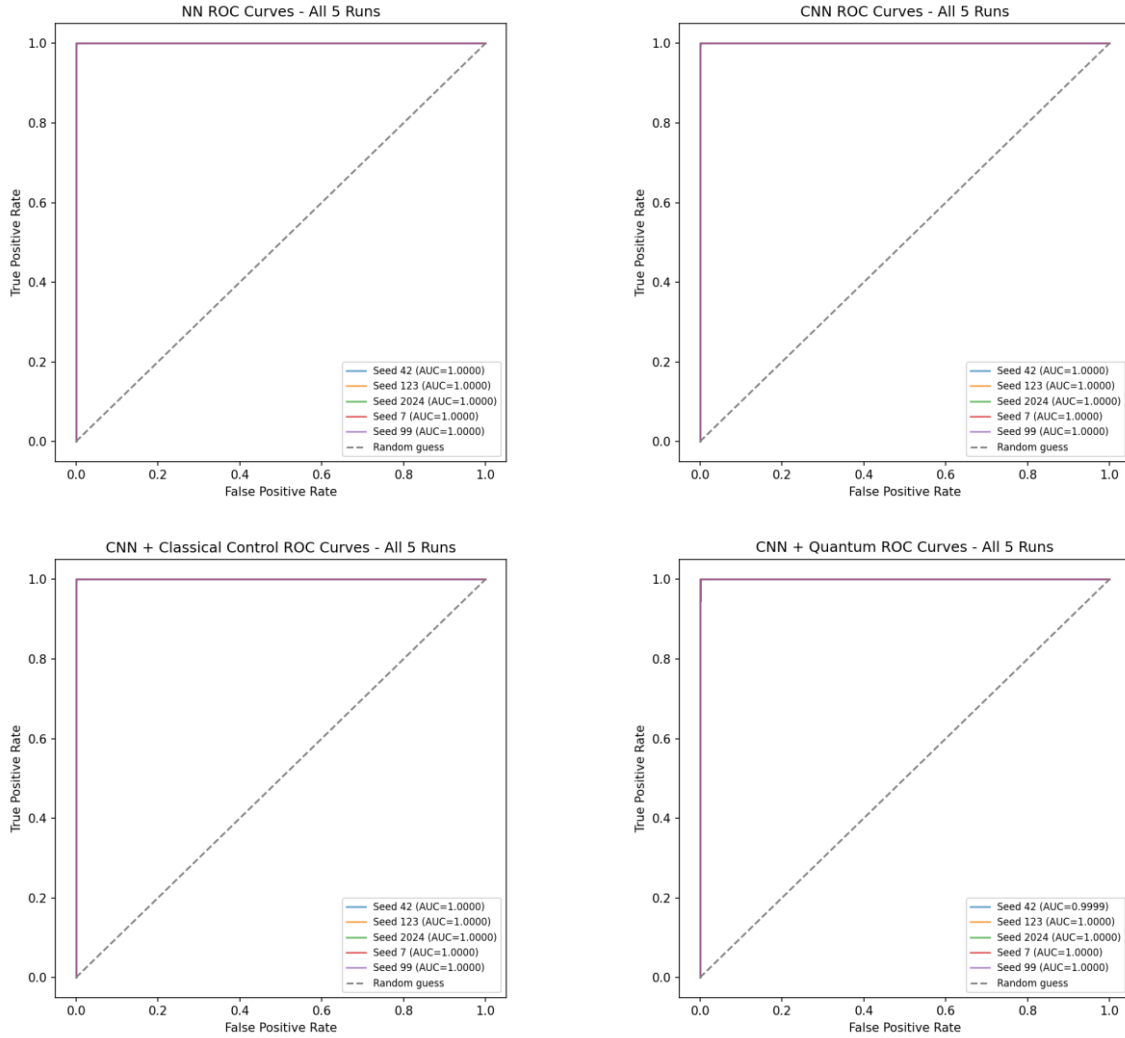


Figure B1-B4. ROC curves for all five runs of NN, CNN, CNN + Classical, and CNN + Quantum (left to right, top to bottom).